Systems of First Order Linear ODEs

1 First Order ODE Systems

1.1 Non-Linear and Linear Systems

The general system of first order equations involving n dependent variables defined on an interval $I \subseteq \mathbb{R}$ can be written as

$$\begin{cases} y_1'(t) = F_1(t, y_1, \ldots, y_n) \\ y_2'(t) = F_2(t, y_1, \ldots, y_n) \\ \qquad \vdots \\ y_n'(t) = F_n(t, y_1, \ldots, y_n) \end{cases} \qquad \text{for} \quad t \in I. \tag{1}$$

$$\text{with initial conditions} \quad y_1(t_0) = x_1, \quad y_2(t_0) = x_2, \quad \ldots, \quad y_n(t_0) = x_n,$$

The solution to the system can be expressed in vector form as

$$\boldsymbol{y}(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{pmatrix}$$

Theorem 1.1 (Existence and Uniqueness)**.**

For the system of first order ODEs (1), if the functions $F_1, F_2, \ldots, F_n$ and the partial derivatives $\dfrac{\partial F_i}{\partial y_j}$ for all pairs $i, j = 1, 2, \ldots, n$ are continuous on a region R defined by

$$R := [t_0 - a, t_0 + a] \times [x_1 - b, x_1 + b] \times \cdots \times [x_n - b, x_n + b] \subseteq \mathbb{R}^{n+1},$$

where constants $a, b > 0$, then there exists a unique solution $\boldsymbol{y}(t)$ for $t \in (t_0 - h, t_0 + h)$ to the IVP, where

$$h = \min\{a, b/M\} \quad \text{and} \quad M = \max_{\boldsymbol{v} \in R}\{|F_1(\boldsymbol{v})|, |F_2(\boldsymbol{v})|, \ldots, |F_n(\boldsymbol{v})|\}.$$

Any n-th order ODE can be rewritten as a system of n first order ODEs. Given an n-th order ODE

$$y^{(n)} = F(t, y, y', \ldots, y^{(n-1)}),$$

setting $z_1 = y, z_2 = y', \ldots, z_n = y^{(n-1)}$, we obtain the system

$$\begin{cases} z_1' = z_2 \\ z_2' = z_3 \\ \qquad \vdots \\ z_{n-1}' = z_n \\ z_n' = F(t, z_1, z_2, \ldots, z_n) \end{cases}$$

which is a system of n first order ODEs.

Definition 1.1 (First-Order Linear System).

A system of first order ODEs is called **linear** if it can be expressed in the form

$$\boldsymbol{y}'(t) = \boldsymbol{P}(t)\boldsymbol{y}(t) + \boldsymbol{g}(t) \qquad \text{for} \quad t \in I, \tag{2}$$

where

$$\boldsymbol{y}(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{pmatrix} \qquad \boldsymbol{g}(t) = \begin{pmatrix} g_1(t) \\ g_2(t) \\ \vdots \\ g_n(t) \end{pmatrix} \qquad \boldsymbol{P}(t) = \begin{pmatrix} p_{11}(t) & p_{12}(t) & \cdots & p_{1n}(t) \\ p_{21}(t) & p_{22}(t) & \cdots & p_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1}(t) & p_{n2}(t) & \cdots & p_{nn}(t) \end{pmatrix}$$

Definition 1.2 (Homogeneous First-Order Linear System).

A first order linear system is called **homogeneous** if $\boldsymbol{g}(t) = \boldsymbol{0}_n$ for all $t \in I$, i.e., it can be expressed as

$$\boldsymbol{y}'(t) = \boldsymbol{P}(t)\boldsymbol{y}(t) \qquad \text{for} \quad t \in I.$$

Theorem 1.2 (Existence and Uniqueness for First-Order Linear Systems).

For the first-order linear system (2), if all entries of $\boldsymbol{P}(t)$ and $\boldsymbol{g}(t)$ are continuous on an interval I, then for any $t_0 \in I$ and $\boldsymbol{y}_0 \in \mathbb{R}^n$, there exists a unique solution $\boldsymbol{y}(t)$ to the IVP

$$\begin{cases} \boldsymbol{y}'(t) = \boldsymbol{P}(t)\boldsymbol{y}(t) + \boldsymbol{g}(t) \\ \boldsymbol{y}(t_0) = \boldsymbol{a} \in \mathbb{R}^n \end{cases} \qquad \text{for} \quad t \in I.$$

1.2 Homogeneous Linear Systems

In this section, we will focus on the following homogeneous system defined on an open interval $I \subseteq \mathbb{R}$:

$$\boldsymbol{y}'(t) = \boldsymbol{P}(t)\boldsymbol{y}(t) \quad \text{for} \quad t \in I, \tag{3}$$

where $\boldsymbol{y} \in C^1(I)^n$ and $\boldsymbol{P} \in C(I)^{n\times n}$, i.e., all entries of $\boldsymbol{P}$ and $\boldsymbol{y}$ are continuous on I.

Theorem 1.3 (Principle of Superposition).

If $\boldsymbol{y}_1(t)$ and $\boldsymbol{y}_2(t)$ are solutions to the homogeneous first-order linear system (3), then the linear combination

$$\boldsymbol{y}(t) = c_1\boldsymbol{y}_1(t) + c_2\boldsymbol{y}_2(t)$$

is also a solution for any constants c_1, c_2.

Definition 1.3 (Wronskian).

The **Wronskian** of n vector functions $\boldsymbol{f}_1(t), \boldsymbol{f}_2(t), \ldots, \boldsymbol{f}_n(t) \in C(I)^n$ is defined as

$$W(\boldsymbol{f}_1, \boldsymbol{f}_2, \ldots, \boldsymbol{f}_n)(t) \quad := \quad \det\left(\begin{bmatrix} | & & | \\ \boldsymbol{f}_1(t) & \cdots & \boldsymbol{f}_n(t) \\ | & & | \end{bmatrix}\right) \quad \text{for} \quad t \in I.$$

Remark. The Wronskian for vector functions is different from the Wronskian for scalar functions. There is no derivative involved in the definition of the Wronskian for vector functions.

For any $t \in I$,

$$W(\boldsymbol{f}_1, \boldsymbol{f}_2, \dots, \boldsymbol{f}_n)(t) \neq 0 \quad \iff \quad \boldsymbol{f}_1(t), \boldsymbol{f}_2(t), \dots, \boldsymbol{f}_n(t) \text{ are linearly independent.}$$

Theorem 1.4 (Liouville's Formula)**.**

Let $\boldsymbol{y}_1(t), \boldsymbol{y}_2(t), \dots, \boldsymbol{y}_n(t)$ be n solutions to the homogeneous system (3). Then the Wronskian of these solutions is given by

$$W(\boldsymbol{y}_1, \boldsymbol{y}_2, \dots, \boldsymbol{y}_n)(t) = c \cdot \exp\left(\int \operatorname{tr}(\boldsymbol{P}(t))\, \mathrm{d}t\right) \quad \text{for} \quad t \in I,$$

where c is a constant not depending on t.

Remark. The **trace** of a square matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ is defined as the sum of its diagonal entries:

$$\operatorname{tr}(\boldsymbol{A}) := a_{11} + a_{22} + \cdots + a_{nn}.$$

Proof (Liouville's Formula)**.**

Before the proof, we need to introduce several theorems from linear algebra:

1. For any square matrix $\boldsymbol{A}$,
$$\frac{\mathrm{d}}{\mathrm{d}t}(\det(\boldsymbol{A})) = \operatorname{tr}\left(\operatorname{adj}(\boldsymbol{A})\frac{\mathrm{d}\boldsymbol{A}}{\mathrm{d}t}\right). \tag{4}$$

2. The trace remains unchanged under cycle permutations:
$$\operatorname{tr}(\boldsymbol{ABC}) = \operatorname{tr}(\boldsymbol{BCA}). \tag{5}$$

3. Property of the adjugate matrix:
$$\operatorname{adj}(\boldsymbol{A})\boldsymbol{A} = \boldsymbol{A}\operatorname{adj}(\boldsymbol{A}) = \det(\boldsymbol{A})\boldsymbol{I}. \tag{6}$$

Now, let $\boldsymbol{Y} := \begin{bmatrix} \boldsymbol{y}_1 & \boldsymbol{y}_2 & \cdots & \boldsymbol{y}_n \end{bmatrix} \in C^1(I)^{n\times n}$ and $W := \det(\boldsymbol{Y})$ be the Wronskian. Then we have

$$\begin{aligned} W' &= \operatorname{tr}(\operatorname{adj}(\boldsymbol{Y})\boldsymbol{Y}') \\ &= \operatorname{tr}(\operatorname{adj}(\boldsymbol{Y})\boldsymbol{PY}) \qquad \text{since} \quad \boldsymbol{y}' = \boldsymbol{Py} \\ &= \operatorname{tr}(\boldsymbol{Y}\operatorname{adj}(\boldsymbol{Y})\boldsymbol{P}) \\ &= \operatorname{tr}(W\boldsymbol{IP}) \\ &= W\operatorname{tr}(\boldsymbol{P}). \end{aligned}$$

Hence the Liouville's formula. □

Theorem 1.5.

If $\boldsymbol{y}_1, \boldsymbol{y}_2, \dots, \boldsymbol{y}_n$ are n solutions to a homogeneous system (3), then

$$\forall t \in I, \quad W(\boldsymbol{y}_1, \boldsymbol{y}_2, \dots, \boldsymbol{y}_n)(t) = 0 \qquad \iff \qquad \exists t_0 \in I \text{ s.t. } W(\boldsymbol{y}_1, \boldsymbol{y}_2, \dots, \boldsymbol{y}_n)(t_0) = 0.$$

***Proof*.**

Proven by Liouville's formula. □

Theorem 1.6.

If the n solutions $\boldsymbol{y}_1(t), \boldsymbol{y}_2(t), \dots, \boldsymbol{y}_n(t)$ of the homogeneous system (3) are **linearly independent** for each $t \in I$, then every solution of the system can be expressed as a linear combination:

$$\boldsymbol{y}(t) = c_1\boldsymbol{y}_1(t) + c_2\boldsymbol{y}_2(t) + \cdots + c_n\boldsymbol{y}_n(t) \quad \text{for} \quad t \in I$$

in **exactly one way**.

Remark. The Principle of Superposition indicates the solutions form a vector space, thus there exist bases. Therefore, the essence of this theorem is to show that the dimension of the solution space is exactly n.

***Proof*.**

Let $\boldsymbol{\phi}(t)$ be any solution to the homogeneous system and let $\boldsymbol{Y} := \begin{bmatrix} \boldsymbol{y}_1 & \boldsymbol{y}_2 & \cdots & \boldsymbol{y}_n \end{bmatrix}$. We pick some $t_0 \in I$ and define the constant vector $\boldsymbol{\xi} := \boldsymbol{\phi}(t_0) \in \mathbb{R}^n$. Since $\boldsymbol{y}_1(t), \dots, \boldsymbol{y}_n(t)$ are linearly independent for any $t \in I$, we have $\det(\boldsymbol{Y})(t_0) \neq 0$. Then the following equation has a unique solution $\boldsymbol{c} \in \mathbb{R}^n$:

$$\boldsymbol{Y}(t_0) \cdot \boldsymbol{c} = \boldsymbol{\xi} \quad \text{for} \quad \boldsymbol{c} \in \mathbb{R}^n.$$

Now we define the linear combination of $\boldsymbol{y}_1, \boldsymbol{y}_2, \dots, \boldsymbol{y}_n$ as a new vector function:

$$\boldsymbol{\eta}(t) := \boldsymbol{Y}(t) \cdot \boldsymbol{c} \quad \text{for} \quad t \in I.$$

Construct the following IVP: the homogeneous system (3) with the initial condition $\boldsymbol{y}(t_0) = \boldsymbol{\xi}$. Since $\boldsymbol{\phi}(t)$ and $\boldsymbol{\eta}(t)$ are both solutions to this IVP, by the uniqueness of solutions (Theorem 1.2), we have $\boldsymbol{\phi}(t) = \boldsymbol{\eta}(t)$ for all $t \in I$. Thus any solution $\boldsymbol{\phi}(t)$ can be expressed as a linear combination of the fundamental solutions. □

Definition 1.4 (Fundamental Set of Solutions)**.**

A set of n **linearly independent** solutions $\{\boldsymbol{y}_1(t), \boldsymbol{y}_2(t), \dots, \boldsymbol{y}_n(t)\}$ to the homogeneous system (3) is called a **fundamental set of solutions**.

Definition 1.5 (Fundamental Matrix)**.**

A matrix function $\boldsymbol{Y}(t) \in C^1(I)^{n\times n}$ is called a **fundamental matrix** of the homogeneous system (3) if its columns form a fundamental set of solutions.

For a square matrix $\boldsymbol{Y}$ whose columns are solutions to the homogeneous system (3), we have

$$\det(\boldsymbol{Y}) \neq 0 \quad \Longleftrightarrow \quad \boldsymbol{Y} \text{ is a fundamental matrix.}$$

Theorem 1.7 (Existence of Fundamental Matrix)**.**

For the homogeneous system (3), there exists at least one fundamental matrix $\boldsymbol{Y}(t)$.

***Proof*.**

We need to show that there exist at least n linearly independent solutions. We can obtain the solutions with Theorem 1.2 by constructing n IVPs with initial conditions at the same point t_0 but different values.

For the i-th IVP, we set the initial condition as $\boldsymbol{y}(t_0) = \boldsymbol{e}_i$, where $\boldsymbol{e}_i$ is the i-th standard basis vector. Then $\boldsymbol{Y}(t_0) = \boldsymbol{I}_n$. Since the Wronskian $W(\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n)(t_0) = \det(\boldsymbol{Y})(t_0) = 1 \neq 0$, the solutions are linearly independent throughout I by Theorem 1.5. □

Theorem 1.8.

Consider the homogeneous system (3) with $\boldsymbol{P} \in C(I, \mathbb{R}^{n\times n})$. Suppose $\boldsymbol{u}, \boldsymbol{v} \in C(I, \mathbb{R}^n)$ are two vector functions such that $\boldsymbol{y} = \boldsymbol{u} + \mathrm{i}\boldsymbol{v} \in C(I, \mathbb{C}^n)$ is a solution to the system. Then both $\boldsymbol{u}$ and $\boldsymbol{v}$ are solutions to the system.

2 Homogeneous Systems with Constant Coefficients

In this section, we will focus on the following system defined on an open interval $I \subseteq \mathbb{R}$:

$$\boldsymbol{y}'(t) = \boldsymbol{A} \cdot \boldsymbol{y}(t) \quad \text{for} \quad t \in I, \tag{7}$$

where $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ is a constant matrix. We will introduce two methods to solve this system:

1. The method of matrix exponentials.

2. The method of eigenvalues and eigenvectors.

2.1 Method 1: Matrix Exponentials

2.1.1 Matrix Exponentials

Theorem 2.1.

For any square matrices $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{n\times n}$,

$$\boldsymbol{AB} = \boldsymbol{BA} \quad \Longrightarrow \quad e^{\boldsymbol{A}+\boldsymbol{B}} = e^{\boldsymbol{A}} e^{\boldsymbol{B}}.$$

***Proof*.**

Using the Binomial Theorem. □

Theorem 2.2.

For any square matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, we have $\det\left(e^{\boldsymbol{A}}\right) \neq 0$, and $\left(e^{\boldsymbol{A}}\right)^{-1} = e^{-\boldsymbol{A}}$.

***Proof*.**

Let $\boldsymbol{A}$ be an arbitrary square matrix. Since $\boldsymbol{A}(-\boldsymbol{A}) = -\boldsymbol{A}^2 = (-\boldsymbol{A})\boldsymbol{A}$, we have $e^{\boldsymbol{A}} e^{-\boldsymbol{A}} = e^{\boldsymbol{A}-\boldsymbol{A}} = e^{\boldsymbol{O}} = \boldsymbol{I}$. □

Theorem 2.3.

For a constant square matrix $\boldsymbol{A}$:

$$\frac{\mathrm{d}(e^{\boldsymbol{A}t})}{\mathrm{d}t} = \boldsymbol{A} e^{\boldsymbol{A}t} = e^{\boldsymbol{A}t} \boldsymbol{A}$$

Proof.

$$\frac{\mathrm{d}(e^{\boldsymbol{A}t})}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\sum_{k=0}^{\infty}\frac{\boldsymbol{A}^k t^k}{k!} = \boldsymbol{A}\sum_{k=1}^{\infty}\frac{\boldsymbol{A}^{k-1}t^{k-1}}{(k-1)!} = \boldsymbol{A}e^{\boldsymbol{A}t} = \sum_{k=1}^{\infty}\frac{\boldsymbol{A}^{k-1}t^{k-1}}{(k-1)!}\cdot\boldsymbol{A} = e^{\boldsymbol{A}t}\boldsymbol{A}$$

□

Theorem 2.3 indicates that the columns of the matrix $\boldsymbol{Y}(t) := e^{\boldsymbol{A}t}$ are solutions to the homogeneous system (7), and Theorem 2.2 shows its invertibility. Thus, $\boldsymbol{Y}(t)$ is a **fundamental matrix**. Then, the **general solution** to the system can be expressed as

$$\boldsymbol{y}(t) = e^{\boldsymbol{A}t}\cdot\boldsymbol{c} \quad \text{for} \quad t \in I,$$

where $\boldsymbol{c} \in \mathbb{R}^n$ is an arbitrary constant vector.

2.1.2 Computing Matrix Exponentials

Now, how to compute $e^{\boldsymbol{A}t}$?

Definition 2.1 (Semisimple Matrix)**.**

A square matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ is called **semisimple** if it is **diagonalizable**, i.e., there exists an invertible matrix $\boldsymbol{P}$ such that $\boldsymbol{P}^{-1}\boldsymbol{A}\boldsymbol{P} = \boldsymbol{D}$, where $\boldsymbol{D}$ is a diagonal matrix.

Remark. The diagonal entries of $\boldsymbol{D}$ are the eigenvalues of $\boldsymbol{A}$.

Definition 2.2 (Nilpotent Matrix)**.**

A square matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ is called **nilpotent** if there exists a positive integer k such that $\boldsymbol{A}^k = \boldsymbol{O}$.

Theorem 2.4.

For a diagonalizable matrix $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{\Lambda}\boldsymbol{P}^{-1} \in \mathbb{R}^{n\times n}$, we have

$$e^{\boldsymbol{A}} = \boldsymbol{P}e^{\boldsymbol{\Lambda}}\boldsymbol{P}^{-1} = \boldsymbol{P}\begin{pmatrix} e^{\lambda_1} & & \\ & \ddots & \\ & & e^{\lambda_n}\end{pmatrix}\boldsymbol{P}^{-1} \qquad \text{where } \lambda_1, \ldots, \lambda_n \text{ are eigenvalues of } \boldsymbol{A}.$$

Theorem 2.5 (S-N Decomposition)**.**

For any square matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, there **uniquely** exist a semisimple matrix $\boldsymbol{S}$ and a nilpotent matrix $\boldsymbol{N}$ such that

$$\boldsymbol{A} = \boldsymbol{S} + \boldsymbol{N} \quad \text{and} \quad \boldsymbol{S}\boldsymbol{N} = \boldsymbol{N}\boldsymbol{S}.$$

Additionally, $\boldsymbol{S}$ shares the same eigenvalues as $\boldsymbol{A}$ with the same algebraic multiplicities.

When computing $e^{\boldsymbol{A}t}$, there are two cases: $\boldsymbol{A}$ is diagonalizable or not. If $\boldsymbol{A}$ is diagonalizable, we can compute $e^{\boldsymbol{A}t}$ with Theorem 2.4. If $\boldsymbol{A}$ is not diagonalizable, we then apply the S-N Decomposition to $\boldsymbol{A}$. Since $\boldsymbol{S}$ and $\boldsymbol{N}$ commute, we have

$$e^{\boldsymbol{A}t} = e^{\boldsymbol{S}t+\boldsymbol{N}t} = e^{\boldsymbol{S}t}e^{\boldsymbol{N}t}.$$

Because $\boldsymbol{S}$ is semisimple, we can compute $e^{\boldsymbol{S}t}$ with Theorem 2.4. For the nilpotent matrix $\boldsymbol{N}$, since $\boldsymbol{N}^k = \boldsymbol{O}$ for some k, the series expansion of $e^{\boldsymbol{N}t}$ terminates within dinitely many terms:

$$e^{\boldsymbol{N}t} = \boldsymbol{I} + \boldsymbol{N}t + \frac{\boldsymbol{N}^2 t^2}{2!} + \cdots + \frac{\boldsymbol{N}^{k-1} t^{k-1}}{(k-1)!}.$$

Thus it is computable.

2.2 Method 2: Eigenvalues and Eigenvectors

Definition 2.3 (Generalized Eigenvector).

For a square matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ and an eigenvalue λ of $\boldsymbol{A}$, a nonzero vector $\boldsymbol{\xi} \in \mathbb{R}^n$ is called a **generalized eigenvector** of $\boldsymbol{A}$ corresponding to λ if there exists a positive integer k such that

$$(\boldsymbol{A} - \lambda \boldsymbol{I})^k \boldsymbol{\xi} = \boldsymbol{0}.$$

If k is the least integer satisfying the above equation, then $\boldsymbol{\xi}$ is called a generalized eigenvector of order k.

Theorem 2.6 (Generalized Eigenspace).

The **generalized eigenspace** of an eigenvalue λ of $\boldsymbol{A}$ with algebraic multiplicity m, denoted by G_λ, is given by

$$G_\lambda = \ker\left((\boldsymbol{A} - \lambda \boldsymbol{I})^m\right) \quad \text{and} \quad \dim(G_\lambda) = m.$$

Consider the homogeneous system (7). Suppose $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ has k distinct eigenvalues:

$$\det(x\boldsymbol{I} - \boldsymbol{A}) = (x-\lambda_1)^{m_1}(x-\lambda_2)^{m_2}\cdots(x-\lambda_k)^{m_k},$$

where m_i is the algebraic multiplicity of λ_i and $\sum_{i=1}^k m_i = n$. For each eigenvalue λ_i, we need to find m_i linearly independent solutions: $\boldsymbol{y}_1^{(i)}, \boldsymbol{y}_2^{(i)}, \dots, \boldsymbol{y}_{m_i}^{(i)}$, so that we have n linearly independent solutions in total.

For each λ_i, whose geometric multiplicity is q_i, there are two cases:

1. $1 \leqslant q_i = m_i \leqslant n$, there are m_i L.I. eigenvectors;
2. $1 \leqslant q_i < m_i \leqslant n$.

Case 1

For the first case, we can construct the solutions for λ_i as

$$\begin{aligned}
\boldsymbol{y}_1^{(i)}(t) &= \boldsymbol{r}_1^{(i)} e^{\lambda_i t}, \\
\boldsymbol{y}_2^{(i)}(t) &= \boldsymbol{r}_2^{(i)} e^{\lambda_i t}, \\
&\vdots \\
\boldsymbol{y}_{m_i}^{(i)}(t) &= \boldsymbol{r}_{m_i}^{(i)} e^{\lambda_i t}.
\end{aligned}$$

where $\boldsymbol{r}_1^{(i)}, \boldsymbol{r}_2^{(i)}, \dots, \boldsymbol{r}_{m_i}^{(i)}$ are m_i L.I. eigenvectors corresponding to λ_i.

Case 2

By Theorem 2.6, there are m_i L.I. generalized eigenvectors for λ_i: $\boldsymbol{\xi}_1^{(i)}, \boldsymbol{\xi}_2^{(i)}, \ldots, \boldsymbol{\xi}_{m_i}^{(i)}$. We use each of them to construct a solution. For the j-th generalized eigenvector $\boldsymbol{\xi}_j^{(i)}$, we first construct the following m_i vectors:

$$\begin{aligned}
\boldsymbol{\eta}_0^{(i,j)} &:= \boldsymbol{\xi}_j^{(i)}, \\
\boldsymbol{\eta}_1^{(i,j)} &:= (\boldsymbol{A} - \lambda_i \boldsymbol{I})\boldsymbol{\xi}_j^{(i)}, \\
\boldsymbol{\eta}_2^{(i,j)} &:= (\boldsymbol{A} - \lambda_i \boldsymbol{I})^2\boldsymbol{\xi}_j^{(i)}, \\
&\vdots \\
\boldsymbol{\eta}_{m_i-1}^{(i,j)} &:= (\boldsymbol{A} - \lambda_i \boldsymbol{I})^{m_i-1}\boldsymbol{\xi}_j^{(i)}.
\end{aligned}$$

Remark. Some of these vectors may be zero. If $\boldsymbol{\xi}_j^{(i)}$ is of order s $(1 \leqslant s \leqslant m_i)$, then $\boldsymbol{\eta}_s^{(i,j)} = \boldsymbol{\eta}_{s+1}^{(i,j)} = \cdots = \boldsymbol{0}$, and $\boldsymbol{\eta}_{s-1}^{(i,j)}$ is an ordinary eigenvector, while the rest $s-1$ nonzero vectors are generalized eigenvectors (nonzero).

Now we construct the solution as:

$$\boldsymbol{y}_j^{(i)}(t) = \left(\boldsymbol{\eta}_0^{(i,j)} + \boldsymbol{\eta}_1^{(i,j)} t + \boldsymbol{\eta}_2^{(i,j)} \frac{t^2}{2!} + \cdots + \boldsymbol{\eta}_{m_i-1}^{(i,j)} \frac{t^{m_i-1}}{(m_i-1)!}\right) \cdot e^{\lambda_i t} \qquad \text{for} \quad j = 1, 2, \ldots, m_i.$$

It can be verified that $\boldsymbol{y}_j^{(i)}$ is indeed a solution to the system by direct substitution: $\dfrac{\mathrm{d}}{\mathrm{d}t}\left(\boldsymbol{y}_j^{(i)}\right) = \boldsymbol{A} \cdot \boldsymbol{y}_j^{(i)}$.

The General Solution

Finally, we can express the general solution to the system as

$$\begin{aligned}
\boldsymbol{y}(t) &= \sum_{i=1}^{k} \sum_{j=1}^{m_i} c_j^{(i)} \boldsymbol{y}_j^{(i)}(t) \\
&= \sum_{i=1}^{k} \sum_{j=1}^{m_i} c_j^{(i)} \left(\boldsymbol{\eta}_0^{(i,j)} + \boldsymbol{\eta}_1^{(i,j)} t + \boldsymbol{\eta}_2^{(i,j)} \frac{t^2}{2!} + \cdots + \boldsymbol{\eta}_{m_i-1}^{(i,j)} \frac{t^{m_i-1}}{(m_i-1)!}\right) \cdot e^{\lambda_i t} \\
&= \sum_{i=1}^{k} \sum_{j=1}^{m_i} c_j^{(i)} e^{\lambda_i t} \sum_{s=0}^{m_i-1} \boldsymbol{\eta}_s^{(i,j)} \frac{t^s}{s!}
\end{aligned}$$

where k is the number of distinct eigenvalues, m_i is the algebraic multiplicity of λ_i, and $c_j^{(i)}$ are arbitrary constants.

Relation with the Matrix Exponential The above solution can be rewritten as

$$\begin{aligned}
\boldsymbol{y}(t) &= \sum_{i=1}^{k} \sum_{j=1}^{m_i} c_j^{(i)} e^{\lambda_i t} \sum_{s=0}^{m_i-1} \boldsymbol{\eta}_s^{(i,j)} \frac{t^s}{s!} \\
&= \sum_{i=1}^{k} \sum_{j=1}^{m_i} c_j^{(i)} e^{\lambda_i t} \left[\sum_{s=0}^{m_i-1} (\boldsymbol{A} - \lambda_i \boldsymbol{I})^s \frac{t^s}{s!}\right] \boldsymbol{\xi}_j^{(i)} \\
&= \sum_{i=1}^{k} \sum_{j=1}^{m_i} c_j^{(i)} e^{\lambda_i t} \left[\sum_{s=0}^{\infty} (\boldsymbol{A} - \lambda_i \boldsymbol{I})^s \frac{t^s}{s!}\right] \boldsymbol{\xi}_j^{(i)} \\
&= e^{\boldsymbol{A}t} \sum_{i=1}^{k} \sum_{j=1}^{m_i} c_j^{(i)} \boldsymbol{\xi}_j^{(i)}
\end{aligned}$$

Note that for each eigenvalue λ_i, $\boldsymbol{\xi}_1^{(i)}, \boldsymbol{\xi}_2^{(i)}, \ldots, \boldsymbol{\xi}_{m_i}^{(i)}$ are m_i L.I. generalized eigenvectors we picked, whose existence is guaranteed by Theorem 2.6. Additionally, generalized eigenvectors corresponding to different eigenvalues are also linearly independent. Thus, the span of all these n generalized eigenvectors is exactly the whole space $\mathbb{R}^n$. We can rewrite the linear combination of them into a single arbitrary constant vector $\boldsymbol{c} \in \mathbb{R}^n$, which gives the same result as the matrix exponential method.

3 Non-Homogeneous Linear Systems

3.1 Method of Undetermined Coefficients

3.2 Variation of Parameters

Consider the non-homogeneous linear system

$$\boldsymbol{y}'(t) = \boldsymbol{P}(t)\boldsymbol{y}(t) + \mathbf{g}(t) \qquad \text{for} \quad t \in I, \tag{8}$$

where $\boldsymbol{P} \in C(I)^{n\times n}$ and $\mathbf{g} \in C(I)^{n}$, and the corresponding homogeneous system is

$$\boldsymbol{y}'(t) = \boldsymbol{P}(t)\boldsymbol{y}(t) \qquad \text{for} \quad t \in I. \tag{9}$$

If we already have a fundamental matrix, then we can construct a particular solution using Variation of Parameters.

Theorem 3.1 (Variation of Parameters)**.**

Consider the non-homogeneous system (8). Suppose $\boldsymbol{F}(t)$ is a **fundamental matrix** of the corresponding homogeneous system (9). Then a **particular solution** to system (8) can be found by setting

$$\boldsymbol{y}_p(t) = \boldsymbol{F}(t) \cdot \boldsymbol{u}(t),$$

where $\boldsymbol{u}(t)$ is given by

$$\boldsymbol{u}(t) = \int \boldsymbol{F}(t)^{-1}\mathbf{g}(t)\,\mathrm{d}t\,.$$

Thus the **general solution** to system (8) is

$$\begin{aligned} \boldsymbol{y}(t) &= \boldsymbol{y}_h(t) + \boldsymbol{y}_p(t) \\ &= \boldsymbol{F}(t) \cdot (\boldsymbol{c} + \boldsymbol{u}(t)) \\ &= \boldsymbol{F}(t) \cdot \boldsymbol{c} + \boldsymbol{F}(t) \int \boldsymbol{F}(t)^{-1}\mathbf{g}(t)\,\mathrm{d}t \qquad \text{for} \quad t \in I, \end{aligned}$$

where $\boldsymbol{c} \in \mathbb{R}^n$ is an arbitrary constant vector.